

Proof Suppose, for contradiction, that the statement P is false. Then [present some argument which leads to a contradiction of some sort].

Hence our assumption that P is false must be false. Thus P is true as required. $\square$

Axiom 5.1.1 (The induction principle) Suppose that $P(n)$ is a statement involving a general positive integer n . Then $P(n)$ is true for all positive integers $1, 2, 3, \dots$ if

- (i) $P(1)$ is true, and
- (ii) $P(k) \Rightarrow P(k+1)$ for all positive integers k .

Proof We use induction on n .

Base case: [Prove the statement $P(n_0)$]

Inductive step: Suppose now as inductive hypothesis that [$P(k)$ is true] for some integer k such that $k \geq n_0$. Then [deduce that $P(k+1)$ is true]. This proves the inductive step.

Conclusion: Hence, by induction, [$P(n)$ is true] for all integers $n \geq n_0$. $\square$

$\mathbb{Z}$ denotes the set of all integers.†

$\mathbb{Z}^+$ denotes the set of all positive integers (natural‡ or counting numbers): $1, 2, 3$, etc.

$\mathbb{Z}^{\geq}$ denotes the set of all non-negative integers: $0, 1, 2, 3$, etc.

$\mathbb{Q}$ denotes the set of all rational numbers (fractions).

$\mathbb{R}$ denotes the set of all real numbers (i.e. numbers expressible as infinite decimals).

$\mathbb{R}^+$ denotes the set of all positive real numbers.

$\mathbb{R}^{\geq}$ denotes the set of all non-negative real numbers.

$\mathbb{C}$ denotes the set of all complex numbers.

Definition 6.1.1 Two sets A and B are [equal], written $A = B$, if they have precisely the same elements, i.e. $A = B$ means $x \in A \Leftrightarrow x \in B$.

Definition 6.1.4 Given sets A and B we say that A is a subset of B , written $A \subseteq B$, or $B \supseteq A$, when every element of A is an element of B , i.e. $x \in A \Rightarrow x \in B$. If A and B are in addition unequal so that B contains some element not contained in A , then we say that A is a [proper] subset of B and write $A \subset B$.†

Thus

$$A = B \Leftrightarrow (A \subseteq B \text{ and } B \subseteq A).$$

Theorem 6.3.4 Let A, B and C be subsets of some universal set U (i.e. $A, B, C \in \mathcal{P}(U)$). Then we have the following identities.

- (i) associativity: $A \cup (B \cap C) = (A \cup B) \cap C$, $A \cap (B \cup C) = (A \cap B) \cup C$.
- (ii) commutativity: $A \cup B = B \cup A$, $A \cap B = B \cap A$.
- (iii) distributivity: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
- (iv) De Morgan laws:† $(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$.
- (v) complementation: $A \cup A^c = U$, $A \cap A^c = \emptyset$.
- (vi) double complement: $(A^c)^c = A$.

Definition 8.4.1 Given a function $f: X \rightarrow Y$, the subset of the codomain Y consisting of those elements which are values of f is called the [image] of f and is denoted by $\text{Im} f$. Thus

$$\text{Im} f = \{ f(x) \mid x \in X \}.$$

Definition 9.1.1 Suppose that $f: X \rightarrow Y$ is a function.

(i) If no element of Y is assigned to more than one element of X , i.e. the function takes a different value for each point of the domain, then we say that the function f is an [injection] (or that it is injective or one-to-one). In symbols we can write this

$$\forall x_1, x_2 \in X, (x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2))$$

or equivalently using the contrapositive

$$\forall x_1, x_2 \in X, (f(x_1) = f(x_2) \Rightarrow x_1 = x_2).$$

(ii) If each element of Y is assigned to some element of X , i.e. each point of the codomain is a value of the function, then we say that the function f is a [surjection] (or that it is surjective or onto). In symbols we can write this

$$\forall y \in Y, \exists x \in X, y = f(x).$$

(iii) If f is both an injection and a surjection then we say that it is a [bijection] (or that it is bijective or one-to-one and onto).

We can reformulate these definitions using the notion of a pre-image.

Definition 9.1.2 For a function $f: X \rightarrow Y$, given an element $y \in Y$ a [pre-image] of y (under f) is an element $x \in X$ such that $y = f(x)$.

Theorem 9.2.3 Let $f: X \rightarrow Y$. Then f is invertible if and only if it is a bijection. Furthermore, if it is invertible, then its inverse function is unique.

Theorem 10.2.1 (The addition principle) Suppose that X and Y are disjoint finite sets. Then $X \cup Y$ is finite and $|X \cup Y| = |X| + |Y|$.

Theorem 10.2.3 (The multiplication principle) Suppose that X and Y are finite sets such that $|X| = n$ and $|Y| = m$. Then the Cartesian product $X \times Y$ is a finite set and $|X \times Y| = mn$.

Proposition 10.3.1 Suppose that X and Y are (not necessarily disjoint) finite sets. Then $X \cup Y$ is a finite set and

$$|X \cup Y| = |X| + |Y| - |X \cap Y|.$$

Proposition 10.3.2 Suppose that X, Y and Z are finite sets. Then $X \cup Y \cup Z$ is a finite set and

$$|X \cup Y \cup Z| = |X| + |Y| + |Z| - |X \cap Y| - |X \cap Z| - |Y \cap Z| + |X \cap Y \cap Z|.$$

Definition 11.3.1 Let a and b be two integers, at least one of which is non-zero. Then the [greatest common divisor] or highest common factor of a and b is the unique positive integer d such that

- (i) d is a common divisor, i.e. d divides a and d divides b ,
- (ii) d is greater than every other common divisor:

$$c \text{ divides } a \text{ and } c \text{ divides } b \Rightarrow c \leq d.$$

Definition 11.3.2 Two integers a and b , not both zero, are called [coprime] or relatively prime when $(a, b) = 1$, in other words their only common factors are 1 and -1 .

Proposition 11.3.4 An integer a is odd if and only if $a = 2q + 1$ for some integer q .

Proposition 12.1.2 Suppose that X and Y are non-empty finite sets with $|X| = m$ and $|Y| = n$. Then the number of functions $X \rightarrow Y$ is given by n^m .

Proposition 12.1.4 Suppose that X and Y are non-empty sets with $|X| = m$ and $|Y| = n$. Then the number of injections $X \rightarrow Y$ is given by $n(n-1)\dots(n-m+1)$.

Theorem 15.1.1 (The division theorem) Let a and b be integers with $b > 0$. Then there are unique integers q and r such that

$$a = bq + r \quad \text{and} \quad 0 \leq r < b.$$

Proposition 15.2.3 If $n \in \mathbb{Z}^+$ is a perfect square (i.e. the square of an integer) then $n = 3q$ or $n = 3q + 1$ for some $q \in \mathbb{Z}$.

Lemma 16.1.1 If a positive integer b divides a then b must be the greatest common divisor of a and b , i.e. $\gcd(a, b) = b$.

Lemma 16.1.2 For non-zero integers a and b suppose that

$$a = bq + r \quad \text{where } q, r \in \mathbb{Z}.$$

Then $\gcd(a, b) = \gcd(b, r)$.

Example 16.2.3 To find the greatest common divisor of 232 and 136.

The Euclidean algorithm gives the following.

$$\begin{array}{lll} \text{Step 1:} & 232 & = 136 \times 1 + 96 \\ \text{Step 2:} & 136 & = 96 \times 1 + 40 \\ \text{Step 3:} & 96 & = 40 \times 2 + 16 \\ \text{Step 4:} & 40 & = 16 \times 2 + 8 \\ \text{Step 5:} & 16 & = 8 \times 2 + 0 \end{array}$$

Hence $(232, 136) = 8$.

Theorem 17.1.1 Let a, b be integers at least one of which is non-zero with greatest common divisor $\gcd(a, b)$. Then there exist integers m and n such that

$$\gcd(a, b) = am + bn.$$

Corollary 17.2.1 Let $d = \gcd(a, b)$ be the greatest common divisor of the integers a and b . Then c is a common divisor of a and b if and only if c divides d , i.e. $D(a, b) = D(d)$.

Proposition 17.3.1 Two non-zero integers a and b are coprime if and only if there exist integers m and n such that

$$am + bn = 1.$$

Theorem 17.3.2 Suppose that a, b and c are positive integers with a and b coprime. Then

$$a \text{ divides } bc \Rightarrow a \text{ divides } c.$$

Theorem 18.2.1 For positive integers a, b and c , there exist integers m and n such that

$$am + bn = c$$

if and only if the greatest common divisor $\gcd(a, b)$ divides c .

Proposition 18.3.2 Suppose that a and b are coprime non-zero integers. Then, for $(m, n) \in \mathbb{Z}^2$,

$$am + bn = 0 \Leftrightarrow (m, n) = (bq, -aq) \text{ for some } q \in \mathbb{Z}.$$

Proposition 18.4.1 Suppose that $(m, n) = (m_0, n_0) \in \mathbb{Z}^2$ is a solution to the diophantine equation $am + bn = c$, i.e. $am_0 + bn_0 = c$. Then, for $(m, n) \in \mathbb{Z}^2$,

$$am + bn = c \Leftrightarrow a(m - m_0) + b(n - n_0) = 0.$$

Definition 19.1.1 Two integers a and b are congruent modulo m when $a - b$ is divisible by m , i.e. there exists an integer q such that $a - b = mq$. We write $a \equiv b \pmod{m}$. If the modulus m is clear we may sometimes omit it and just write $a \equiv b$.

Proposition 19.1.2

- (i) Reflexive property. For all integers a , $a \equiv a \pmod{m}$.
- (ii) Symmetric property. If a and b are integers such that $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$.
- (iii) Transitive property. If a, b and c are integers such that $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$.

Proposition 19.1.3 (Modular arithmetic) Suppose that a_1, a_2, b_1 and b_2 are integers such that $a_1 \equiv a_2 \pmod{m}$ and $b_1 \equiv b_2 \pmod{m}$. Then

- (i) $a_1 + b_1 \equiv a_2 + b_2 \pmod{m}$,
- (ii) $a_1 - b_1 \equiv a_2 - b_2 \pmod{m}$,
- (iii) $a_1 b_1 \equiv a_2 b_2 \pmod{m}$.

Proposition 19.2.4 Two integers a and b are congruent modulo m if and only if $r_m(a) = r_m(b)$.

Proposition 19.3.1 Suppose that a is an integer which divides m (i.e. m/a is an integer). Then

$$ab_1 \equiv ab_2 \pmod{m} \Leftrightarrow b_1 \equiv b_2 \pmod{m/a}.$$

Proposition 19.3.2 Suppose that a is an integer such that a and m are coprime (i.e. the greatest common divisor $(a, m) = 1$). Then

$$ab_1 \equiv ab_2 \pmod{m} \Leftrightarrow b_1 \equiv b_2 \pmod{m}.$$

Problem 20.0.1 Given a positive integer m and integers a and b , solve the congruence

$$ax \equiv b \pmod{m},$$

i.e. find all integers x which satisfy this congruence.

Theorem 20.1.5 Let m be a positive integer. Suppose that a and b are integers such that a and m are coprime. Then the linear congruence

$$ax \equiv b \pmod{m}$$

has a solution. This solution is unique up to congruence modulo m .

Theorem 20.1.7 The linear congruence $ax \equiv b \pmod{m}$ has a solution if and only if the greatest common divisor (a, m) divides b . In this case the number of different solutions modulo m is $\gcd(a, m)$.

Proposition 20.2.1 For integers a, b , and positive integers m , there is a bijection

$$f: \{(x, y) \in \mathbb{Z}^2 \mid ax + my = b\} \rightarrow \{x \in \mathbb{Z} \mid ax \equiv b \pmod{m}\}$$

given by $f(x, y) = x$.

$$\gcd(a, b) = am + bn$$

$$(a, b) = 1 \iff \exists m, n \in \mathbb{Z} \text{ such that } am + bn = 1$$

$$(a, b) = 1 \text{ and } a \mid bc \implies a \mid c$$

$$c \mid a \text{ and } c \mid b \iff c \mid \gcd(a, b)$$

$$\gcd(a, b, c) = \gcd(\gcd(a, b), c)$$

A particular solution is just one specific integer pair (x_0, y_0) satisfying

$$ax_0 + by_0 = c.$$

How to find it

1. Compute $d = \gcd(a, b)$
2. Check $d \mid c$
3. Express d as $am + bn$
4. Multiply by c/d

Then all integer solutions are:

$$x = x_0 + \frac{b}{d}t, \quad y = y_0 - \frac{a}{d}t, \quad t \in \mathbb{Z}.$$

For

$$ax + by = 0,$$

let $d = \gcd(a, b)$. Then all solutions are:

$$x = \frac{b}{d}t, \qquad y = -\frac{a}{d}t, \qquad t \in \mathbb{Z}.$$

Solving Linear Congruences:

If $d \mid a$, $d \mid b$, and $d \mid m$, then divide the whole congruence by d :

$$ax \equiv b \pmod{m} \iff \frac{a}{d}x \equiv \frac{b}{d} \pmod{\frac{m}{d}}.$$

Then continue until the coefficient of x is coprime to the modulus.

Solve

$$ax + my = b.$$

Any solution (x, y) gives a solution x to the congruence.

Problems III #12.

Suppose that there is an injection $f : \mathbb{Z}^+ \rightarrow X$.

Prove by contradiction that X is a finite set.

Solution. Suppose, for a contradiction, that X is a finite set.

Let $n = |X|$, which is a nonnegative integer.

Define $g : \mathbb{N}_{n+1} \rightarrow X$ by $g(x) = f(x)$.

That is, g is the restriction of f from its original domain $\mathbb{Z}^+$ to the new domain $\mathbb{N}_{n+1}$.

Since the codomain $\mathbb{N}_1$ contains only one element, 1, we must have that $f(x_1) = 1$ and $f(x_2) = 1$. Thus

$f(x_1) = f(x_2)$, which contradicts f being injective, since $x_1 \neq x_2$.

Thus X has at most 1 element.

This implies that X is finite with cardinality either 0 (in which case X is empty) or 1. In either case, $|X| \leq 1$.

This completess the proof of the base case.

Problems III, #9.

Prove Proposition 11.1.4: If $X \rightarrow \mathbb{N}_n$ is an injection, then X is finite and $|X| \leq n$.

Solution.

Base case. $n = 1$. Suppose that $f : X \rightarrow \mathbb{N}_1$ is an injection.

Suppose for a contradiction that X has at least two distinct elements.

Call these elements x_1 and x_2 .

Since the codomain $\mathbb{N}_1$ contains only one element, 1, we must have that $f(x_1) = 1$ and $f(x_2) = 1$. Thus

$f(x_1) = f(x_2)$, which contradicts f being injective, since $x_1 \neq x_2$.

Thus X has at most 1 element.

This implies that X is finite with cardinality either 0 (in which case X is empty) or 1. In either case, $|X| \leq 1$.

This completess the proof of the base case.

Inductive step. Suppose n is a positive integer with the property that the statement "If $X \rightarrow \mathbb{N}_n$ is an

injection, then X is finite and $|X| \leq n$." is true.

We want to prove that this statement with n replaced by $n + 1$ is true.

To this end:

Suppose $f : X \rightarrow \mathbb{N}_{n+1}$ is an injection.

We want to show that X is finite and $|X| \leq n + 1$.

Again, we want to be able to use the "previous level",

We separate the proof into cases:

Case 1. $n + 1$ is not in the image of f .

Then the function $g : X \rightarrow \mathbb{N}_n$ defined by $g(x) = f(x)$ is well-defined.

Claim. g is an injection.

To see the claim, which is easy, suppose $g(x_1) = g(x_2)$. By definition, this means $f(x_1) = f(x_2)$. Since f is

an injection, this implies $x_1 = x_2$.

We conclude that g is an injection.

Now we can apply the inductive hypothesis to conclude that X is finite and $|X| \leq n \leq n + 1$.

Case 2. $n + 1$ is in the image of f .

Then there exists $x_0 \in X$ such that $f(x_0) = n + 1$.

Claim. The function $g : X - \{x_0\} \rightarrow \mathbb{N}_n$ defined by $g(x) = f(x)$ is well-defined and an injection.

To see the claim, we first show that g is well-defined. Since $g(x) = f(x)$ and since f has codomain $\mathbb{N}_{n+1}$,

the only thing that could go wrong is that $g(x_1) = n + 1$ for some $x_1 \in X - \{x_0\}$. But then this implies that

$f(x_0) = n + 1 = g(x_1) = f(x_1)$. But $x_0 \in X - \{x_0\}$, so that $x_1 \neq x_0$. Thus we have a contradiction to f

being an injection.

Next, we see that g is an injection. This is easy. Suppose $g(x_1) = g(x_2)$ for some $x_1, x_2 \in X - \{x_0\}$. Then

$f(x_1) = g(x_1) = g(x_2) = f(x_2)$. Since f is an injection, we conclude that $x_1 = x_2$. This proves that g is an

injection.

Now we can apply the inductive hypothesis to conclude that $X - \{x_0\}$ is a finite set and $|X - \{x_0\}| \leq n$.

Thus, by the addition principle, $X = (X - \{x_0\}) \cup \{x_0\}$ is a finite set and

$|X| = |X - \{x_0\}| + |\{x_0\}| \leq n + 1$. This proves the inductive conclusion.

By induction we have proved the statement we wanted to prove.

Solution. Suppose, for a contradiction, that there exists a smallest real number.

Call this number x . Then $\frac{x}{2} = \frac{1}{2}x$ is a real number since $\frac{1}{2}$ is a real number and the set of real numbers is

closed under multiplication.

Since $\frac{1}{2} > 0$ and $x > 0$, we have that $\frac{x}{2} = \frac{1}{2}x$ is positive.

We claim that $\frac{x}{2} < x$. Indeed, this inequality is **equivalent** to $x = 2\frac{x}{2} < 2x$ since $2 > 0$ and by an axiom in the

book (you don't need to remember that is it Axiom 3.1.2). Subtracting x from both sides, we see that this

inequality is equivalent to $0 = x - x < 2x - x$, which is true since x is positive.

We have obtained contradiction to x being the smallest positive real number. The proof is complete.

11.1. Suppose that $\mathbb{N}_n \rightarrow X$ is a surjection. Prove, by induction on n , that X is a finite set and that $|X| \leq n$.

Solution.

Base case. $n = 1$. Suppose $f : \mathbb{N}_1 \rightarrow X$ is a surjection.

We claim that X , which must be nonempty since $\mathbb{N}_1$ is, is finite with cardinality 1, which solves the base

case.

To see the claim, assume for a contradiction that X has two or more elements. Let x_1 and x_2 be distinct

elements of X . Since f is surjective and since the there is only one possible input 1, we must have that

$f(1) = x_1$ and $f(2) = x_1$. This contradicts f being a well-defined function.

Inductive Step. Suppose n is a positive integer with the property that whenever $\mathbb{N}_n \rightarrow X$ is a surjection,

then X is a finite set and that $|X| \leq n$.

We want to prove this property for n replaced by $n + 1$.

So we have to prove an implication.

Suppose $f : \mathbb{N}_{n+1} \rightarrow X$ is a surjection.

We want to show that then X is a finite set and that $|X| \leq n + 1$.

But to do this we need to relate the hypothesis to the previous level n .

We do this by cases, with the first case the easiest.

Remove $n + 1$ from the domain to define a new function. That is, define

$g : \mathbb{N}_n \rightarrow X$ by $g(x) = f(x)$ for $x \in \mathbb{N}_n \subseteq \mathbb{N}_{n+1}$.

Case 1. g is a surjection. In this case, since $g : \mathbb{N}_n \rightarrow X$ is a surjection, we can use the inductive

hypothesis to conclude that X is a finite set and that $|X| \leq n \leq n + 1$. So we are done in this case.

Case 2. g is not a surjection. Then there exists $x_0 \in X$ such that x_0 is not equal to $g(i) = f(i)$ for any

$i \in \mathbb{N} + n$. But $x_0 = f(j)$ for some $j \in \mathbb{N}_{n+1}$. So it must be that $f(n + 1) = x_0$.

Claim. $h : \mathbb{N}_n \rightarrow X - \{x_0\}$ defined by $h(x) = f(x)$ is a well-defined surjection.

To see that h is well-defined, we simply observe that the only element of $\mathbb{N}_{n+1}$ that maps by f to x_0 is the

element $n + 1$.

So $f(x) \in X - \{x_0\}$ for all $x \in \mathbb{N}_n$. This implies that h is well-defined.

To see that h is a surjection, we argue as follows:

Let $x \in X - \{x_0\}$. Since f is a surjection, there exists $y \in \mathbb{N}_{n+1}$ such that $f(y) = x$.

Now $y \neq n + 1$ since $f(n + 1) = x_0 \notin X - \{x_0\}$.

Therefore $y \in \mathbb{N}_n$.

This in turn implies that $h(y) = f(y) = x$

This proves that $h : \mathbb{N}_n \rightarrow X - \{x_0\}$ is a surjection.

But we are in the "previous level". So, by the inductive hypothesis, we have that $X - \{x_0\}$ is finite and

$|X - \{x_0\}| \leq n$.

We are almost done.

We have that $X = (X - \{x_0\}) \cup \{x_0\}$ is a disjoint union of finite sets.

Thus X is finite and $|X| = |X - \{x_0\}| + |\{x_0\}|$. That is,

$|X| = |X - \{x_0\}| + 1$, i.e., $|X - \{x_0\}| = |X| - 1$.

Combining this with the above, we obtain

$|X| - 1 \leq n$. This implies $|X| \leq n + 1$.

We have proved the inductive conclusion.

We conclude by induction the desired result.

Prove by induction that for any positive integer n ,

$$2^n + 2 \cdot 2^n + \cdots + n \cdot 2^n = \frac{n(n+1)}{2} 2^n.$$

Answer:

Base case: $n = 1$. The left-hand side is $2^1 = 2$.

The right-hand side is $\frac{1(1+1)}{2} 2^1 = \frac{2}{2} 2 = 2$.

Since the LHS equals the RHS, the base case is true.

Indutive step: Suppose a positive integer n has the property that

$$\begin{aligned} 2^n + 2 \cdot 2^n + \cdots + n \cdot 2^n &= \frac{n(n+1)}{2} 2^n \text{ (this is the inductive hypothesis). We compute that} \\ 2^{n+1} + 2 \cdot 2^{n+1} + \cdots + (n+1) \cdot 2^{n+1} &= (2^{n+1} + 2 \cdot 2^{n+1} + \cdots + n \cdot 2^{n+1}) + (n+1) \cdot 2^{n+1} \\ &= 2 \cdot (2^n + 2 \cdot 2^n + \cdots + n \cdot 2^n) + (n+1) \cdot 2^{n+1} \\ &= 2 \frac{n(n+1)}{2} 2^n + (n+1) 2^{n+1} \\ &= \frac{(n+2)(n+1)}{2} 2^{n+1} \\ &= \frac{(n+1)((n+1)+1)}{2} 2^{n+1}. \end{aligned}$$

This proves the formula for the $n + 1$ case. By induction we are done.

The universal set is $\mathbb{R}$. Prove or disprove: $\forall x \neq 0, \exists y > 0, y < x^2$.

Solution. The statement is **True**.

Proof. Let x be a nonzero real number. Define $y = \frac{x^2}{2}$. Since $x \neq 0$ we have $|x| > 0$, and hence $x^2 = |x|^2 > 0$. Let $y = \frac{x^2}{2}$.

Since $x^2 > 0$ and $\frac{1}{2} > 0$, we have that their product $y = \frac{x^2}{2}$ is positive.

Moreover, by an axiom in the book, the inequality $y = \frac{x^2}{2} < x^2$ is equivalent to the inequality $x^2 < 2x^2$

obtained by multiplying both sides by the positive number 2. This last inequality is seen to be true by

subtracting x^2 from both sides to yield the equivalent inequality $0 = x^2 - x^2 < 2x^2 - x^2 = x^2$, which we have

proved above. This completes the proof.

Problem 1 (Review Proposition 5.2.1). Prove by induction on n that, for all integers n such that $n \geq 4$, we have the inequality

$$n^2 \leq 2^n.$$

Solution. We use induction on n .

Base case: For $n = 4$,

$$4^2 = 16 \quad \text{and} \quad 2^4 = 16,$$

so $4^2 \leq 2^4$.

Inductive step: Assume that for some integer $k \geq 4$, $k^2 \leq 2^k$.

We must show that: $(k+1)^2 \leq 2^{k+1}$.

From the inductive hypothesis, $2k^2 \leq 2^{k+1}$.

So it is enough to prove that $(k+1)^2 \leq 2k^2$. Now

$$(k+1)^2 \leq 2k^2 \iff k^2 + 2k + 1 \leq 2k^2 \iff 2k + 1 \leq k^2.$$

Since $k \geq 4$, we have

$$k^2 \geq 4k \geq 2k + 2 > 2k + 1.$$

Hence

$$(k+1)^2 \leq 2k^2 \leq 2^{k+1}.$$

This proves the inductive step.

Therefore, by induction,

$$n^2 \leq 2^n \quad \text{for all integers } n \geq 4.$$

Problem 2 (Exercise 17.6). Suppose that a_1, a_2, b_1, b_2 are positive integers such that a_1 and b_1 are coprime and a_2 and b_2 are coprime. Prove that

$$a_1 b_2 = a_2 b_1 \implies a_1 = a_2 \text{ and } b_1 = b_2.$$

Solution. Assume that

$$a_1 b_2 = a_2 b_1.$$

We will show that $a_1 = a_2$ and $b_1 = b_2$.

Since

$$a_1 b_2 = a_2 b_1,$$

it follows that a_1 divides $a_2 b_1$. Because a_1 and b_1 are coprime, Euclid's lemma implies that a_1 divides a_2 .

Similarly, from

$$a_2 b_1 = a_1 b_2,$$

we see that a_2 divides $a_1 b_2$. Because a_2 and b_2 are coprime, Euclid's lemma implies that a_2 divides a_1 .

Thus $a_1 \mid a_2$ and $a_2 \mid a_1$. Since both are positive integers, we conclude that $a_1 = a_2$.

Substituting into the original equation gives

$$a_1 b_2 = a_1 b_1.$$

Since $a_1 > 0$, we may cancel a_1 and obtain $b_2 = b_1$.

Hence

$$a_1 = a_2 \quad \text{and} \quad b_1 = b_2.$$

Problem 3 (Set theory practice). Let A, B, C be sets. Suppose

$$A \cup C \subseteq B \cup C \quad \text{and} \quad A \cap C \subseteq B \cap C.$$

Prove that $A \subseteq B$.

Solution. We must show that every element of A belongs to B .

Let $x \in A$. We consider two cases.

Case 1: $x \in C$.

Then $x \in A \cap C$. By the hypothesis

$$A \cap C \subseteq B \cap C,$$

we obtain $x \in B \cap C$, and therefore $x \in B$.

Case 2: $x \notin C$.

Since $x \in A$, certainly $x \in A \cup C$. By the hypothesis

$$A \cup C \subseteq B \cup C,$$

we obtain $x \in B \cup C$. But $x \notin C$, so it must be that $x \in B$.

In either case, $x \in B$. Therefore every element of A belongs to B , so

$$A \subseteq B.$$

Problem 4 (Surjection practice). Let Y be a set and let $f : \mathbb{Z} \rightarrow Y$ be a function satisfying the following properties:

1. f is a surjection,
2. $f(2n) = f(2n+1)$ for all $n \in \mathbb{Z}$.

Let $E \subseteq \mathbb{Z}$ be the set of even integers. Define $g : E \rightarrow Y$, where $g(k) = f(k)$ for $k \in E$. Prove that g is a surjection.

Solution. We must show that every element of Y is the image under g of some element of E .

Let $y \in Y$. Since $f : \mathbb{Z} \rightarrow Y$ is a surjection, there exists some $m \in \mathbb{Z}$ such that

$$f(m) = y.$$

Now either m is even or m is odd.

Case 1: m is even.

Then $m \in E$, and by definition of g ,

$$g(m) = f(m) = y.$$

Case 2: m is odd.

Then $m = 2n + 1$ for some $n \in \mathbb{Z}$. By property (2),

$$f(2n) = f(2n+1) = f(m) = y.$$

Since $2n$ is even, $2n \in E$, and by definition of g ,

$$g(2n) = f(2n) = y.$$

In either case, there exists $k \in E$ such that $g(k) = y$. Therefore g is a surjection.

Problem 5 (Review Quiz 2, Question 2). Let $f : X \rightarrow Y$ be an injection. Suppose that a subset B of Y does not intersect the image of f , that is, there does not exist $x \in X$ such that $f(x) \in B$. Prove that there exists an injection

$$g : X \rightarrow Y - B.$$

In particular:

1. define g ,
2. justify that g is a well-defined function,
3. show that g is an injection.

Solution. Define

$$g : X \rightarrow Y - B \quad \text{by} \quad g(x) = f(x) \quad \text{for all } x \in X.$$

We first show that g is well-defined. Since f is a function, for each $x \in X$ the value $f(x)$ is uniquely determined, so $g(x)$ is uniquely determined as well. Also, by hypothesis there does not exist $x \in X$ such that $f(x) \in B$. Hence for every $x \in X$,

$$f(x) \notin B.$$

Because $f(x) \in Y$, it follows that

$$f(x) \in Y - B.$$

Thus $g(x) = f(x) \in Y - B$ for every $x \in X$, so g is indeed a function from X to $Y - B$.

Now we show that g is injective. Suppose $x_1, x_2 \in X$ and

$$g(x_1) = g(x_2).$$

By definition of g , this means

$$f(x_1) = f(x_2).$$

Since f is injective, we conclude that

$$x_1 = x_2.$$

Therefore g is an injection.

Hence there exists an injection

$$g : X \rightarrow Y - B.$$